



MAX9295A

July 21, 2017

CSI-2 to GMSL2/GMSL1 Serializer

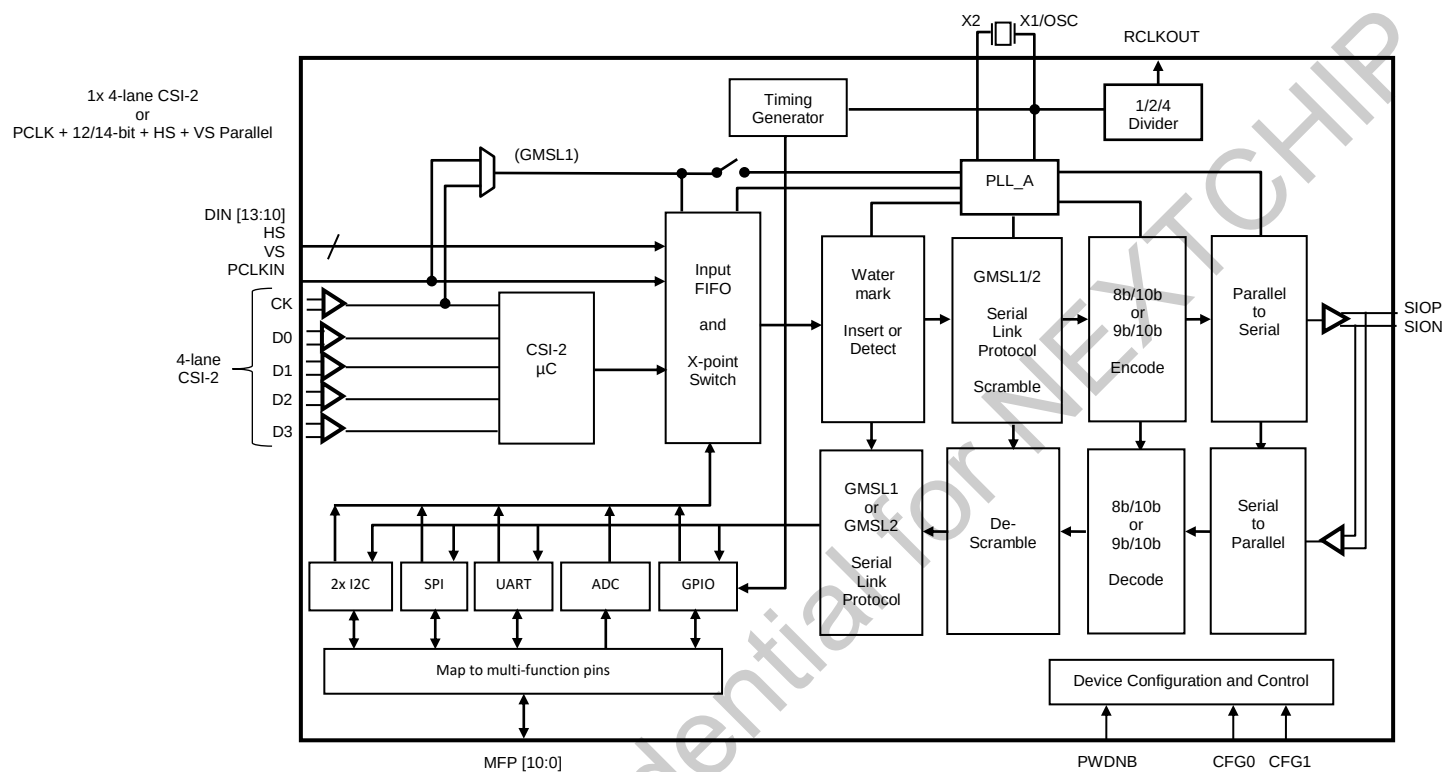
General Description

The MAX9295A converts MIPI CSI-2 or LVCMOS parallel to GMSL2 or GMSL1 while sending and receiving control channel data, enabling transmission of video and bi-directional control data over cables up to 15 meters. GMSL2 operates at fixed rate of 3Gbps or 6Gbps in the forward direction and 187.5Mbps in the reverse direction. In GMSL1 mode, the MAX9295A can be paired with first-generation 3.12Gbps or 1.5Gbps GMSL1 deserializers, or operate up to 4.5Gbps with GMSL2 deserializers in GMSL1 mode. Operation is specified over the automotive temperature range of -40°C to +105°C and the device is AEC-Q100 Grade 2 qualified. Cable can be coax or shielded twisted pair (STP).

Features

- 4-lane MIPI CSI-2 v1.3 port
- MIPI D-PHY v1.2 at 80Mbps - 2.5Gbps per lane
- 150MHz clock parallel input (12/14-bit data + H + V)
- Line fault monitor
- Supports RAW8/10/12/14/16/20, RGB565/666/888, YUV422 8/10-bit, User Defined and Generic Long Packet data types
- 16-channel CSI-2 virtual channel support
- Selectable GMSL2 or backward-compatible GMSL1 mode
- Operates up to 4.5Gbps with GMSL2 deserializers in GMSL1 mode
- CSI-2 ECC and Checksum error detection and flagging
- Watermark generator and checker
- XTAL oscillator or reference input with divided or undivided reference output
- On-chip timing generator for video sync signal generation
- High speed (187.5Mbps) backchannel for reduced-size power-over-cable (POC) LC filter
- Multi-function pins (MFP) programmable as GPO sync output for surround view camera systems
- Multi-function pins (MFP) programmable as I2C, SPI, UART, sampled or register-programmed GPIO
- 8 hardware-selectable device addresses
- 3 Internal and 3 external supply voltage monitor (10-bit ADC)
- Works with 50Ω coax cable or 100Ω STP
- Crosspoint switch remaps input to any desired deserializer output order
- CSI-2 lane polarity flip
- CSI-2 lane reassignment
- Eye opening monitor (EOM) on reverse channel for real-time jitter margin measurement
- Remote/local wake-up from sleep mode
- Software reset
- Concurrent control channel transactions with settable priority level
- Programmable spread spectrum on serial I/O for EMI reduction
- Forward and reverse channel PRBS for BER testing of serial link
- 32-bit CRC protection of video data
- 16-bit CRC protection of control channel data
- Retransmission of control channel data upon CRC error detection
- 5mm x 5mm TQFN or TQFN side-wettable package with 0.5mm pitch
- 1.7V to 1.9V analog supply
- Core supply accepts +/-5% tolerance supply with nominal value of 1.3V, 1.2V, 1.1V or 1.0V
- 1.7V to 3.6V I/O supply
- ±8kV Contact and ±15kV air ISO10605 and IEC61000-4-2 ESD tolerance on serial I/O
- -40°C to +105°C operating temperature
- Meets AEC-Q100 Grade 2 specification

MAX9295A Block Diagram



Absolute Maximum Ratings

VDD18 to EP*	-0.3V to +2.0V
VDD to EP*	-0.3V to +1.9V
VDDIO to EP*	-0.3V to +3.9V
SIOP, SION to EP*	+0.5V to +VDD18
MFP, PWDNB, Parallel Inputs to EP*	-0.3V to + (V _{VDDIO} + 0.3V)
CSI-2 Inputs to EP*	-0.3V to + (V _{TBD} + 0.3V)
X1/OSC, X2 to EP*	-0.3V to + TBDV
Continuous Power Dissipation, T _A = +70°C	
Multi-Layer Board	
TQFN-SW and TQFN (derate 34.5mW/°C above +70°C)	2758mW
Single-Layer Board	
TQFN-SW and TQFN (derate 21.3mW/°C above +70°C)	1702mW
Operating Temperature Range	-40°C to +105°C
Junction Temperature	+150°C
Storage Temperature Range	-40°C to +150°C
Soldering temperature (reflow)	+260°C

*EP connected to IC ground.

Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

Package Information

32-pin TQFN-SW (Side-wettable)

Package Code	T3255Y+8
Package Outline Document Number	21-100156
Land Pattern Document Number	90-0013
Single-Layer Board	
Junction-to-Ambient Thermal Resistance (θ _{JA})	47°C/W
Junction-to-Case Thermal Resistance (θ _{JC})	1.7°C/W
Four-Layer Board	
Junction-to-Ambient Thermal Resistance (θ _{JA})	29°C/W
Junction-to-Case Thermal Resistance (θ _{JC})	1.7°C/W

32-pin TQFN

Package Code	T3255+8
Package Outline Document Number	21-0140
Land Pattern Document Number	90-0013
Single-Layer Board	
Junction-to-Ambient Thermal Resistance (θ _{JA})	47°C/W
Junction-to-Case Thermal Resistance (θ _{JC})	1.7°C/W
Four-Layer Board	
Junction-to-Ambient Thermal Resistance (θ _{JA})	29°C/W
Junction-to-Case Thermal Resistance (θ _{JC})	1.7°C/W

Recommended Operating Conditions

			MIN	TYP	MAX	Unit
	VDD18		1.7	1.8	1.9	V
	VDD	1.0V	0.95	1.0	1.05	V
		1.1V	1.045	1.1	1.155	V
		1.2V	1.14	1.2	1.26	V
		1.3V	1.235	1.3	1.365	V
	VDDIO	1.8V	1.7	1.8	1.9	V
		3.3V	3.0	3.3	3.6	V
	VDD18			25		mVp-p
	VDD	1.0V		25		mVp-p
		1.1V		50		mVp-p
		1.2V		50		mVp-p
		1.3V		50		mVp-p
	VDDIO	1.8V		50		mVp-p
		3.3V		100		mVp-p
Operating free-air temperature, T _A			-40	25	105	°C

Note 1: supply noise < 1MHz

DC ELECTRICAL CHARACTERISTICS

(V_{DD18} = 1.7V to 1.9V, V_{DD} = 0.95V to 1.37V, V_{DDIO} = 1.7V to 3.6V, R_L = 100Ω ±1% (differential), T_A = -40°C to +105°C, EP connected to PCB ground, typical values are at, V_{DD18} = V_{DDIO} = 1.8V, V_{DD} = 1.0V, T_A = 25°C, unless otherwise noted.) (Note 1)

Serial Output (SIOP, SION)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Output Voltage Swing	V _{OD}	High drive, GMSL1 or GMSL2, twisted-pair	300	400	500	mV		
		High drive, GMSL1, coax		500		mV		
Change in VOD between Complimentary Output States	ΔV _{OD}				25	mV		
Output Offset Voltage	V _{OS}	(V _{SIOP} + V _{SION}) / 2 = V _{OS}	V _{DD18} - 0.55	V _{DD18} - 0.45	V _{DD18} - 0.35	V		
Change in VOS between Complimentary Output States	ΔV _{OS}				25	mV		
Output Short-Circuit Current	I _{OS}	SIOP or SION = 0V	-60			mA		
Output Short-Circuit Current	I _{OS}	SIOP or SION = 1.9V			25	mA		
Output Termination Resistance (Internal)	R _O	From SIOP or SION to V _{DD18}	50	53	56	Ω		

I/O pins (GPIO)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
High-Level Input Voltage	V _{IH}		V _{DDIO} * 0.7			V		
Low-Level Input Voltage	V _{IL}				V _{DDIO} * 0.3	V		
High-Level Output Voltage	V _{OH}	I _{OH} = -4mA	V _{DDIO} - 0.4			V		
Low-Level Output Voltage	V _{OL}	I _{OL} = 4mA			0.4	V		
Input Current	I _{IN}	V _{IN} = 0 to V _{DDIO} All pull-up/pull-down disabled.			1	μA		

Input Capacitance	C _{IN}	Each pin (note 2)		5		pF		
Input pull-up/ pull-down Resistance	R _{IN}	RS_CFG = 0		40		kΩ		
Input pull-up/ pull-down Resistance	R _{IN}	RS_CFG = 1		1		MΩ		

Open-Drain Pins (SDA₁, SCL₁, SDA₂ RX, TX, SCL₂, LOCK, ERRB/LFLT_B, ODO)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
High-Level Input Voltage	V _{IH2}		V _{DDIO} * 0.7		V _{DDIO}	V		
Low-Level Input Voltage	V _{IL2}				V _{DDIO} * 0.3	V		
Low-Level Open-Drain Output Voltage	V _{OL}	I _{OL} = 4mA			0.4	V		
Input Current	I _{IN}	V _{IN} = 0 to V _{DDIO} . R _{PU} disabled.			1	μA		
Input Capacitance	C _{IN}	Each pin (note 2)		5		pF		
Internal Pull-up Resistor	R _{PU}	RES_CFG = 0		40		kΩ		
Internal Pull-up Resistor	R _{PU}	RES_CFG = 1		1		MΩ		

Input Only Pin (PWDNB)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
High-Level Input Voltage	V _{IH}		V _{DDIO} * 0.7		V _{DDIO}	V		
Low-Level Input Voltage	V _{IL}				V _{DDIO} * 0.3	V		
Input Current	I _{IN}	V _{IN} = 0 to V _{DDIO}			5	μA		
Input Capacitance	C _{IN}	Each pin (note 2)		5		pF		
Internal Pull-down Resistor	R _{PD}			1		MΩ		

LINE FAULT DETECTION INPUT (LMN0, LMN1)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Short-to-Battery to Open Threshold	I _{TB-O0}	Current measured out of LMN0	-25		-15	μA		
Open to Normal Threshold	I _{TO-N0}	Current measured out of LMN0	6.5		9.5	μA		
Normal to Short-to-Ground Threshold	I _{ILG0}	Current measured out of LMN0	16		20	μA		
Open Pin Voltage	V _{IO0}	Pin LMN0		1.25		V		
Short-to-Battery to Open Threshold	I _{TB-O1}	Current measured out of LMN1	-12.5		7.5	μA		
Open to Line-to-Line Threshold	I _{TO-L1}	Current measured out of LMN1	2		4	μA		
Line-to-Line Normal Threshold	I _{IL-N1}	Current measured out of LMN1	5		7	μA		
Normal to Short-to-Ground Threshold	I _{TN-G1}	Current measured out of LMN1	8		10	μA		

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Open Pin Voltage	V_{IO1}	Pin LMN1		0.75		V		

CSI-2 HS Receiver

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Ground shift	V_{GNDSH}		-50		50	mV		
Common-mode voltage HS receive Mode	$V_{CMRX(DC)}$	Note 3, 4	70		330	mV		
Differential input high threshold	V_{IDTH}	Data rate ≤ 1.5 Gbps			70	mV		
		Data rate > 1.5 Gbps			40	mV		
Differential input low threshold	V_{IDTL}	Data rate ≤ 1.5 Gbps	-70			mV		
		Data rate > 1.5 Gbps	-40			mV		
Single-ended input high voltage	V_{IHHS}	Note 3			460	mV		
Single-ended input low voltage	V_{ILHS}	Note 3	-40			mV		
Single-ended threshold for termination enable	$V_{TERM-EN}$				450	mV		
Differential input impedance	Z_{ID}		80	100	125	Ω		
Note: 3. Excluding possible additional RF interference of 100mV peak sine wave beyond 450MHz. 4. This table value includes a ground difference of 50mV between the transmitter and the receiver, the static common-mode level tolerance and variations below 450MHz.								

CSI-2 LP Receiver

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Logic 1 input voltage	V_{IH}	Data rate ≤ 1.5 Gbps	880			mV		
		Data rate > 1.5 Gbps	740			mV		
Logic 0 input voltage, not in ULP State	V_{IL}				550	mV		
Logic 0 input voltage, ULP State	$V_{IL-ULPS}$				300	mV		
Input hysteresis	V_{HYST}		25			mV		
Pin leakage current	I_{LEAK}	-0.05V to 1.35V	-10		10	μA		

Reference Clock Input Requirements (Crystal) (X1, X2)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
X1 Input Capacitance	C_{IN-X1}			7		pF		
X2 Input Capacitance	C_{IN-X2}			13		pF		
Feedback Resistor	R_{FB}			10		k Ω		
Transconductance	g_m			28		mA/V		

Reference Clock Requirements (External Input on X1, X2 floating)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Input High Voltage	V_{IH}	$V_{DDIO} = 1.7V$ to $3.6V$	0.9		$V_{DDIO} + 0.3$	V		
Input Low Voltage	V_{IL}	$V_{DDIO} = 1.7V$ to $3.6V$	-0.3		+0.4	V		
Input Current	I_{INX}	$V_{IN} = 0$ to V_{DD18}	-5		+5	μA		
X1 Input Capacitance	C_{IN-X1}	Note 2		7		pF		
Input Impedance	R_{IN}	Note 2		10		k Ω		

External Resistors Requirements

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
XRES Value	R_{XRES}			402		Ω		

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
XRES Tolerance					±1	%		
R _{PD} Value				49.9		kΩ		
R _{PD} Tolerance					±1	%		
R _{EXT} Value				48.7		kΩ		
R _{EXT} Tolerance					±1	%		

Power Supply (GMSL2 mode)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Supply Current (note 5)	I _{DD1}	V _{DDIO} = 3.3V		10	15	mA		
	I _{DD2}	V _{DD18} = 1.8V		112	125	mA		
	I _{DD3}	V _{DD} = 1.0V		104	180	mA		

ESD Protection

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
SIOP, SION	V _{ESD}	Human Body Model, R _D = 1.5kΩ, C _S = 100pF		±8		kV		
SIOP, SION	V _{ESD}	IEC61000-4-2, R _D = 330Ω, C _S = 150pF, Contact discharge		±8		kV		
SIOP, SION	V _{ESD}	IEC61000-4-2, R _D = 330Ω, C _S = 150pF, Air discharge		±15		kV		
SIOP, SION	V _{ESD}	ISO10605, R _D = 2kΩ, C _S = 330pF, Contact discharge		±8		kV		
SIOP, SION	V _{ESD}	ISO10605, R _D = 2kΩ, C _S = 330pF, Air discharge		±15		kV		
SIOP, SION	V _{ESD}	AEC-Q100-011 Rev-C1, Charged Device Model (CDM)		500		V		
Corner Pins	V _{ESD}	Human Body Model, R _D = 1.5kΩ, C _S = 100pF		±4		kV		
Corner Pins	V _{ESD}	AEC-Q100-011 Rev-C1, Charged Device Model (CDM)		750		V		
All Other Pins	V _{ESD}	Human Body Model, R _D = 1.5kΩ, C _S = 100pF		±4		kV		
All Other Pins	V _{ESD}	AEC-Q100-011 Rev-C1, Charged Device Model (CDM)		500		V		

AC ELECTRICAL CHARACTERISTICS

CSI-2 HS Receiver

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Common-mode interference beyond 450MHz	$\Delta V_{CMRX(HF)}$	Note 2	Data rate ≤ 1.5Gbps		100	mV		
			Data rate > 1.5Gbps		50	mV		
Common-mode interference beyond 50MHz - 450MHz	$\Delta V_{CMRX(LF)}$	Note 1, 4	Data rate ≤ 1.5Gbps	-50	50	mV		
			Data rate > 1.5Gbps	-25	25	mV		
Common-mode termination	C _{CM}	Note 3			60	pF		
Note: 1. Excluding 'static' ground shift of 50mV 2. $\Delta V_{CMRX(HF)}$ is the peak amplitude of a sine wave superimposed on the receiver inputs. 3. For higher bit rates a 14pF capacitor will be needed to meet the common-mode return loss specification. 4. Voltage difference compared to the DC average common-mode potential.								

CSI-2 LP Receiver

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Input pulse rejection	e _{SPIKE}	Note 1, 2, 3			300	V-ps		

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Minimum pulse width response	$T_{\text{MIN-RX}}$	Note 4	20			ns		
Peak interference amplitude	V_{INT}				200	mV		
Interference frequency	f_{INT}		450			MHz		
Note: 1. Time-voltage integration of a spike above V_{IL} when being in LP-0 state or below V_{IH} when being in LP-1 state. eSpike generation will ensure the spike is crossing both $V_{\text{IL,max}}$ and $V_{\text{IH,min}}$ levels. 2. An impulse less than this will not change the receiver state. 3. In addition to the required glitch rejection, implementers shall ensure rejection of known RF-interferers. 4. An impulse greater than this shall toggle the output.								

CSI-2 Data-Clock Timing

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
UI instantaneous	U_{INST}		0.4		12.5	ns		
UI variation	ΔU_{I}	$U_{\text{I}} \geq 1\text{ns}$, within a single burst	-10%		10%	UI		
		$0.667\text{ns} < U_{\text{I}} < 1\text{ns}$, within a single burst	-5%		5%	UI		
Data to Clock Setup Time	$T_{\text{SETUP(RX)}}$	0.08 to 1.0Gbps	0.15			U_{INST}		
		>1.0 to 1.5Gbps	0.20			U_{INST}		
Data to Clock Hold Time	$T_{\text{HOLD(RX)}}$	0.08 to 1.0Gbps	0.15			U_{INST}		
		>1.0 to 1.5Gbps	0.20			U_{INST}		
Static Data to Clock Skew (RX)	$T_{\text{SKEW(RX) Static}}$	>1.5Gbps	-0.20		0.20	U_{INST}		
Dynamic Data to Clock Skew Window RX Tolerance	$T_{\text{SETUP(RX)}} + T_{\text{HOLD(RX) Dynamic}}$	>1.5Gbps	0.50			U_{INST}		

CSI-2 Global Operation Timing

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Timeout for receiver to detect absence of Clock transitions and disable the Clock Lane HS-RX.	$T_{\text{CLK-MISS}}$				60	ns		
Time interval during which the HS receiver ignores any Clock Lane HS transitions, starting from the beginning of $T_{\text{CLK-PREPARE}}$.	$T_{\text{CLK-SETTLE}}$		95		300	ns		
Time for the Clock Lane receiver to enable the HS line termination, starting from the time point when Dn crosses $V_{\text{IL,MAX}}$.	$T_{\text{CLK-TERM-EN}}$		Time for Dn to reach $V_{\text{TERM-EN}}$		38	ns		
Time for the Data Lane receiver to enable the HS line termination, starting from the time point when Dn crosses $V_{\text{IL,MAX}}$.	$T_{\text{D-TERM-EN}}$		Time for Dn to reach $V_{\text{TERM-EN}}$		35ns + $4 \cdot U_{\text{INST}}$	ns		
Time interval during which the HS receiver ignores any Data Lane HS transitions, starting from the beginning of $T_{\text{HS-PREPARE}}$.	$T_{\text{HS-SETTLE}}$		85ns + $6 \cdot U_{\text{INST}}$		145ns + $10 \cdot U_{\text{INST}}$	ns		
Time interval during which the HS-RX ignores any transitions on the Data Lane, following a HS burst.	$T_{\text{HS-SKIP}}$		40		55ns + $4 \cdot U_{\text{INST}}$	ns		

SWITCHING CHARACTERISTICS

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Serial Output Rise/Fall time	$t_{\text{R}}, t_{\text{F}}$	20% to 80%, V_{OD} ; 400mV differential $R_{\text{L}} = 100\Omega$, 500mV single-ended $R_{\text{L}} = 50\Omega$		50	70	ps		
Total Serial Output Jitter (Differential output)	t_{TSOJ1}	Unencoded PRBS7, measured at $V_{\text{OD}} = 0\text{V}$ differential. Forward channel, spread spectrum disabled		0.25		UI		
Deterministic Serial Output Jitter (Differential output)	t_{DSOJ2}	Unencoded PRBS7, measured at $V_{\text{OD}} = 0\text{V}$ differential		0.15		UI		

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Total Serial Output Jitter (Single-ended output)	t_{TSOJ1}	Unencoded PRBS7, measured at $V_O/2$		0.25		UI		
Deterministic Serial Output Jitter (Single-ended output)	t_{DSOJ2}	Unencoded PRBS7, measured at $V_O/2$		0.15		UI		
Serializer Latency	t_{SL}	Time from signal received at CSI-2 input until it appears on GMSL3 output during active video			TBD	T_{PCLK}		
Video Initialization Time	$t_{\text{Link-Start}}$	Time from initial video is applied to CSI-2 until it appears at SIOP, SION. Assumes link is already established.			2	ms		
Lock Time	t_{LOCK}	Time from de-assertion of PWDNB to LOCK = high			7	ms		
GPI-to-GPO Delay	t_{GPD1}	Delay-compensated Mode		10		μS		
GPI-to-GPO Delay	t_{GPD2}	Non-delay-compensated Mode		3		μS		
REF Clock Output Jitter	t_{ROJ}	25MHz divided by 1, 2 or 4. TBD kHz – TBD MHz		TBD		ps		
Timing Generator Output Jitter	t_{TGJ}	Output frequency = 37.125MHz Jitter frequency: TBD kHz – TBD MHz		TBD		ps		
Timing Generator Output Jitter	t_{TGJ}	Output frequency = 27MHz Jitter frequency: TBD kHz – TBD MHz		TBD		ps		

I2C or UART PORT TIMING

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Output Rise Time	t_{R}	30% to 70%, $C_L = 10\text{pF}$ to 100pF , $1\text{k}\Omega$ pullup to VDDIO	20		150	ns		
Output Fall Time	t_{F}	70% to 30%, $C_L = 10\text{pF}$ to 100pF , $1\text{k}\Omega$ pullup to VDDIO	20		150	ns		

I2C Timing

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
SCL Clock Frequency	f_{SCL}	Low f_{SCL} range: (I2CMST_BT = 010, I2CSLV_SH = 10)	9.6		100	kHz		
SCL Clock Frequency	f_{SCL}	Mid f_{SCL} range: (I2CMST_BT 101, I2CSLV_SH = 01)	>100		400	kHz		
SCL Clock Frequency	f_{SCL}	High f_{SCL} range: (I2CMST_BT = 111, I2CSLV_SH = 00)	>400		1000	kHz		
Start Condition Hold Time	$t_{\text{HD:STA}}$	f_{SCL} range, Low	4			μs		
Start Condition Hold Time	$t_{\text{HD:STA}}$	f_{SCL} range, Mid	0.6			μs		
Start Condition Hold Time	$t_{\text{HD:STA}}$	f_{SCL} range, High	0.26			μs		
Low Period of SCL Clock	t_{LOW}	f_{SCL} range, Low	4.7			μs		
Low Period of SCL Clock	t_{LOW}	f_{SCL} range, Mid	1.3			μs		
Low Period of SCL Clock	t_{LOW}	f_{SCL} range, High	0.5			μs		
High Period of SCL Clock	t_{HIGH}	f_{SCL} range, Low	4			μs		
High Period of SCL Clock	t_{HIGH}	f_{SCL} range, Mid	0.6			μs		
High Period of SCL Clock	t_{HIGH}	f_{SCL} range, High	0.26			μs		
Repeated Start Condition Setup Time	$t_{\text{SU:STA}}$	f_{SCL} range, Low	4.7			μs		
Repeated Start Condition Setup Time	$t_{\text{SU:STA}}$	f_{SCL} range, Mid	0.6			μs		
Repeated Start Condition Setup Time	$t_{\text{SU:STA}}$	f_{SCL} range, High	0.26			μs		
Data Hold Time	$t_{\text{HD:DAT}}$	f_{SCL} range, Low	0			ns		

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Data Hold Time	$t_{HD:DAT}$	f_{SCL} range, Mid	0			ns		
Data Hold Time	$t_{HD:DAT}$	f_{SCL} range, High	0			ns		
Data Setup Time	$t_{SU:DAT}$	f_{SCL} range, Low	250			ns		
Data Setup Time	$t_{SU:DAT}$	f_{SCL} range, Mid	100			ns		
Data Setup Time	$t_{SU:DAT}$	f_{SCL} range, High	50			ns		
Setup Time for Stop Condition	$t_{SU:STO}$	f_{SCL} range, Low	4			us		
Setup Time for Stop Condition	$t_{SU:STO}$	f_{SCL} range, Mid	0.6			us		
Setup Time for Stop Condition	$t_{SU:STO}$	f_{SCL} range, High	0.26			us		
Bus Free Time	t_{BUF}	f_{SCL} range, Low	4.7			us		
Bus Free Time	t_{BUF}	f_{SCL} range, Mid	1.3			us		
Bus Free Time	t_{BUF}	f_{SCL} range, High	0.5			us		
Data Valid Time	$t_{VD:DAT}$	f_{SCL} range, Low			3.45	us		
Data Valid Time	$t_{VD:DAT}$	f_{SCL} range, Mid			0.9	us		
Data Valid Time	$t_{VD:DAT}$	f_{SCL} range, High			0.45	us		
Data Valid Acknowledge Time	$t_{VD:ACK}$	f_{SCL} range, Low			3.45	us		
Data Valid Acknowledge Time	$t_{VD:ACK}$	f_{SCL} range, Mid			0.9	us		
Data Valid Acknowledge Time	$t_{VD:ACK}$	f_{SCL} range, High			0.45	us		
Pulse width of spikes suppressed	t_{SP}	f_{SCL} range, Low			50	ns		
Pulse width of spikes suppressed	t_{SP}	f_{SCL} range, Mid			50	ns		
Pulse width of spikes suppressed	t_{SP}	f_{SCL} range, High			50	ns		
Capacitive load each bus line	C_B	Note 2			100	pF		

SPI Master

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Operating Frequency	f_{MCK}				50	MHz		
SCLK period	t_{MCK}			$1/f_{MCK}$		ns		
SCLK Output Pulse-Width High/Low	t_{MCH}, t_{MCL}		$t_{MCK}/2$			ns		
MOSI Output Hold Time After SCLK Rising Edge	t_{MOH}		$t_{MCK}/2$			ns		
MOSI Output Valid to SCLK Rising Edge	t_{MOV}		$t_{MCK}/2$			ns		
MISO Input Valid to SCLK Rising Edge Setup	t_{MIS}		3			ns		
MISO Input to SCLK Rising Edge	t_{MIH}			0		ns		

SPI Slave

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Operating Frequency	f_{MCK}				50	MHz		
SCLK period	t_{MCK}			$1/f_{MCK}$		ns		
SCLK Input Pulse-Width High/Low	t_{MCH}, t_{MCL}		$t_{MCK}/2$			ns		
MISO Output Hold Time After SCLK Rising Edge	t_{MOH}		$t_{MCK}/2$			ns		
MISO Output Valid to SCLK Rising Edge	t_{MOV}		$t_{MCK}/2$			ns		

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
MOSI Input Valid to SCLK Rising Edge Setup	t_{MIS}		3			ns		
MOSI Input to SCLK Rising Edge	t_{MIH}			0		ns		

Parallel Clock Input (PCLK)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Duty Cycle	DC	t_{HIGH}/t_T or t_{LOW}/t_T Note TBD , Figure TBD	35	50	65	%		
Transition Time	t_R, t_F	150MHz, 20% to 80% Note TBD , Figure TBD		1		ns		
		75MHz, 20% to 80% Note TBD , Figure TBD		2		ns		
Jitter	t_J			TBD		ps		

Parallel Input

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Data Setup Time	t_{SET}			1		ns		
Data Hold Time	t_{HOLD}			1		ns		

Reference Clock Input Requirements (Crystal) (X1, X2)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Frequency	f_{XTAL}			25		MHz		
Frequency Stability + Frequency Tolerance	f_{TN}		-200		+200	ppm		

Reference Clock Requirements (External Input on X1, X2 floating)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT	TYPE	GL
Frequency	f_{REF}			25		MHz		
Frequency Stability + Frequency Tolerance	f_{TN}		-200		+200	ppm		
Input Jitter		Forward data rate 6Gbps, Reverse data rate 187.5Mbps, Sinusoidal jitter <1MHz (rising edge)		500	600	ps p-p		
Input Duty Cycle	t_{DUTY}		40		60	%		
Input Rise/Fall Time	t_R, t_F				8	ns		

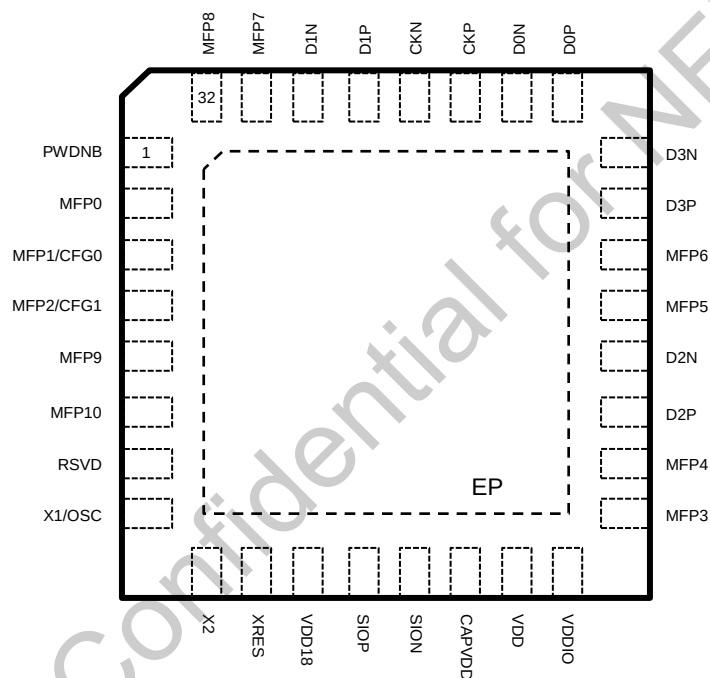
Note 1 Limits are 100% tested at $T_A = +25^\circ\text{C}$ and $T_A = +105^\circ\text{C}$. Limits over the operating temperature range and relevant supply voltage range are guaranteed by design and characterization.

Note 2 Not production tested. Guaranteed by design.

MAX9295A Pin Out

32-lead 5x5mm TQFN and TQFN side-wettable Packages (**Top View** – Exposed Pad and leads on backside)

Configuration pins CFG0 and CFG1 are latched at power-up. CSI-2 default input shown.



9295A Pin Description

Pin	Pin Name	Default Function (CGF1 bit PAR/ CSI-2 = 0)	Default Function (CGF1 bit PAR/ CSI-2 = 1)	Alternate Functions						Description
				A	B	C	D	E	F	
1	PWDN B	PWDNB	PWDNB							PWDNB: Active-low power-down input.
2	MFPO	Hi-Z	PCLK	GPIO0	SCLK	VTG0		CNTL3 (GMSL1)		PCLK: Parallel interface pixel clock input. GPIO0: General-purpose input or output. SCLK: SPI clock output when configured as SPI master. SPI clock input when configured as SPI slave. VTG0: Video timing generator output. CNTL3 (GMSL1): GPI in GMSL1 mode.
3	MFP1/ CFG0	Hi-Z	Hi-Z	GPO1	SS1	BNE	VTG1			CFG0: configuration pin latched at power-up. Connect to a resistor divider ($\pm 1\%$ resistors) between VDDIO and EP. GPO1: General-purpose output. SS1: when configured as SPI master, SS1 is a slave select output. BNE: when configured as SPI slave, BNE output indicates SPI data is available. VTG1: Video timing generator output. Note: Any pin loading in addition to the resistor divider must be high-impedance at power-up. See Table 1 and Table 1 notes for important details.
4	MFP2/ CFG1	Hi-Z	Hi-Z	GPO2	RCLKOUT	VTG2				CFG1: configuration pin latched at power-up. Connect to a resistor divider ($\pm 1\%$ resistors) between VDDIO and EP. GPO2: General purpose output. RCLKOUT: 25MHz frequency reference output. Divide by 2 or 4 available. VTG2: Video timing generator output. Note: Any pin loading in addition to the resistor divider must be high-impedance at power-up. See Table 2 and Table 2 notes for important details.
5	MFP9	SDA_RX	SDA_RX	SDA2	ODO9_ GPI9					SDA_RX: I2C data input/output or UART input. SDA2: pass-through I2C data input/output. ODO9_GPI9: Open-drain output or general-purpose input.
6	MFP10	SCL_TX	SCL_TX	SCL2	ODO10_ GPI10					SCL_TX: I2C clock input/output or UART output. SCL2: pass-through I2C clock input/output. ODO10_GPI10: Open-drain output or general-purpose input.
7	RSVD	RSVD	RSVD							Reserved. Not internally connected. Leave open.
8	X1/OSC	X1/OSC	X1/OSC							Crystal/Oscillator Input. If crystal used, connect to one terminal of a 25MHz ± 200 ppm crystal and connect a load capacitor from X1/OSC to EP (load capacitor value depends on crystal used). If oscillator used, supply a 25MHz ± 200 ppm signal.
9	X2	X2	X2							Crystal Input. Connect to one terminal of a 25MHz ± 200 ppm crystal and connect a load capacitor from X1/OSC to EP (load capacitor value depends on crystal used).
10	XRES	XRES	XRES							Connect a 402 Ω $\pm 1\%$ resistor between XRES and EP.
11	VDD18	VDD18	VDD18							1.7V to 1.9V analog supply. Bypass VDD18 to EP with a 0.1 μ F capacitor placed as close as possible to the device.
12	SIOP	SIOP	SIOP							Non-inverted Coax/Twisted-Pair Serial-Data output.
13	SION	SION	SION							Inverted Twisted-Pair Serial-Data output
14	CAPVDD D	CAPVDD	CAPVDD							When VDD = 1.0V, bypass CAPVDD to EP with 0.1 μ F placed as close as possible to the device. When VDD = 1.3V, 1.2V or 1.1V, bypass CAPVDD to EP with 0.1 μ F and 10 μ F capacitors placed as close as possible to the device, with the smaller-value capacitor closest to CAPVDD.
15	VDD	VDD	VDD							+/-5% tolerance core supply accepts nominal 1.3V, 1.2V, 1.1V or 1.0V. Bypass VDD to EP with a 0.1 μ F capacitor placed as close as possible to the device.
16	VDDIO	VDDIO	VDDIO							1.7V to 3.6V I/O supply. Bypass VDDIO to EP with a 0.1 μ F capacitor placed as close as possible to the device.

17	MFP3	LFLT/ERRB	VS	GPIO3	ADC0	VTG3	LOCK	CNTL2 (GMSL1)		LFLT_ERRB: Line fault or Error output. Goes low when a line fault on LMN0 or LMN1 is detected or a CSI-2 ECC or Checksum error is detected. VS: Parallel interface vertical sync input VTG3: Video timing generator output. GPIO3: General-purpose input or output. ADC0: supply voltage monitor. VTG3: Video timing generator output. LOCK: high indicates PLLs locked with correct serial word boundary alignment. CNTL2 (GMSL1): GPI in GMSL1 mode.
18	MFP4	Hi-Z	HS	RCLKOUT	SS2	RO	VTG4	GPIO4	CNTL1 (GMSL1)	HS: Parallel interface horizontal sync input. RCLKOUT: 25MHz frequency reference output. Divide by 2 or 4 available. SS2: when configured as SPI master, SS2 is a slave select output. RO: when configured as SPI slave, the SPI master can read from local SPI slave when initiating an exchange with the remote slave. RO is an input. VTG4: Video timing generator output. GPIO4: General-purpose input or output. CNTL1 (GMSL1): GPI in GMSL1 mode.
19	D2P	D2P	D0							D2P: CSI-2 data lane 2 non-inverted input. D0: Parallel interface data input.
20	D2N	D2N	D1							D2N: CSI-2 data lane 2 inverted input. D1: Parallel interface data input.
21	MFP5	Hi-Z	D2	ODO5_GPI5	ADC1	LMN0				D2: Parallel interface data input. ODO5_GPI5: Open-drain output or general-purpose input. ADC1: supply voltage monitor. LMN0: line fault monitor input
22	MFP6	Hi-Z	D3	ODO6_GPI6	ADC2	LMN1				D3: Parallel interface data input. ODO6_GPI6: Open-drain output or general-purpose input. ADC2: supply voltage monitor. LMN1: line fault monitor input.
23	D3P	D3P	D4							D3P: CSI-2 data lane 3 non-inverted input. D4: Parallel interface data input.
24	D3N	D3N	D5							D3N: CSI-2 data lane 3 inverted input. D5: Parallel interface data input.
25	D0P	D0P	D6							D0P: CSI-2 data lane 0 non-inverted data input. D6: Parallel interface data input.
26	D0N	D0N	D7							D0N: CSI-2 data lane 0 inverted data input. D7: Parallel interface data input.
27	CKP	CKP	D8							CKP: CSI-2 clock lane non-inverted input. D8: Parallel interface data input.
28	CKN	CKN	D9							CKN: CSI-2 clock lane inverted input. D9: Parallel interface data input.
29	D1P	D1P	D10							D1P: CSI-2 data lane 1 non-inverted data input. D10: Parallel interface data input.
30	D1N	D1N	D11							D1N: CSI-2 data lane 1 inverted data input. D11: Parallel interface data input.
31	MFP7	GPI7	D12	GPIO7	MOSI	SDA1	VTG7	GPO (GMSL1)	CNTL4 (GMSL1)	GPI7: General-purpose input. D12: Parallel interface data input. GPIO7: General-purpose input or output. MOSI: output drives data to an external SPI slave. SDA1: pass-through I2C data input/output. VTG7: Video timing generator output. GPO (GMSL1): General-purpose output in GMSL1 mode CNTL4 (GMSL1): GPI in GMSL1 mode.
32	MFP8	GPO8	D13	GPIO8	MISO	SCL1	VTG8	MS	CNTL0 (GMSL1)	GPO8: General-purpose output. D13: Parallel interface data input. GPIO8: General-purpose input or output. MISO: input receives data from an external SPI slave. SCL1: pass-through I2C clock input/output. VTG8: Video timing generator output. MS: Mode select input selects base mode or bypass mode for control channel.

										CNTL0 (GMSL1): GPI in GMSL1 mode.
EP	EP									Exposed pad. Connect to board ground plane through an array of vias.

GMSL2 Physical Interface

Link Speeds (Forward and Reverse)

Forward transmission on the GMSL2 link is communication from the Serializer to the Deserializer. Reverse transmission is communication from the Deserializer to the Serializer. The GMSL2 link rate is fixed and independent of the video clock. The forward rate is 6Gbps the reverse rate is 187.5Mbps.

Coax/STP Cable Support

Both 50Ω coaxial and 100Ω shielded twisted-pair (STP) cables that meet the GMSL2 channel specification are supported.

Selectable Line Rate

On power up the forward transmission rate is 6Gbps or 3Gbps depending on the configuration pin CFG1. The reverse transmission rate is 187.5Mbps.

Spread Spectrum

Spread spectrum is used to reduce EMI (ElectroMagnetic Interference). The forward channel spread spectrum frequency is programmable in the range of 10-40kHz and the modulation depth is programmable in the range 0-4%. If no forward channel spread spectrum is programmed, up to 4% spread spectrum can be tolerated.

AEQ (Adaptive Equalization)

AEQ automatically adapts the reverse receiver characteristics to match the channel (cable and connectors). This process consists of adjusting echo cancellation and compensating for frequency dependent channel loss. AEQ can be register programmed for frequency of operation, or disabled.

GMSL2 Packet Protocol

A packet-based transmission protocol is employed to efficiently and dynamically use and share full link bandwidth. Link bandwidth is only used by a channel when data is being sent.

9b10b Link Encoding

9b10b is a link encoding scheme that maps 9-bit symbols (512 unique data symbols) and eight additional special characters (K) to 10-bit symbols. This is an efficient encoding scheme that requires significantly less encoding overhead than other variants. To illustrate, this method requires 11% overhead whereas 8b10b encoding requires 25% overhead—effectively apportioning more bandwidth for communications and data transfer. The link is DC-balanced: the running disparity (i.e., the number of 1s and 0s in a given window) is bounded to ± 9 with a maximum run length (i.e., consecutive number of 1s or 0s) limited to seven.

Eight available special K characters are used for link synchronization, as packet delimiters, and other control purposes. The Hamming distance (number of bit differences) in this scheme is at least three between any two special symbols. Therefore, a special symbol cannot transform into another special symbol with one or two bit errors. A cascading sequence of special symbols creates a unique bit pattern nonexistent in any other data symbol combination; this sequence is deployed for symbol and word boundary locking purposes to delimit data packets.

Each packet starts with two words consisting of two special symbols in each word. This packet indicator is repeated for error correction purposes and allows one bit error correction in the receiver for errors that hit packet start words.

NOTE: Reduction of packet start words to one word (two special symbols) to increase bandwidth efficiency is a programmable option. This option should only be employed if the physical link is sufficiently reliable and the additional usable bandwidth is needed. In this scenario, a bit error hitting one of the special symbols in the packet start word will not transform the affected packet into another packet; rather, the packet will be dropped entirely. Dropped control packets can be automatically recovered by the ARQ mechanism (see below). Packets can be protected by an optional 16-bit packet CRC (see below).

Packet Protocols

Below is a description of the different packet protocols.

Sync Words

Sync words are two words sent regularly by the transmitter every 2048 words (default) on each link (Note: the sync word period is a configurable register setting). The transmitter inserts the sync words into the packet stream irrespective of packet boundaries; the receiver removes them to preserve the original packet data stream. These sync words consist of two identical words in both single- or double-packet start mode.

This packet serves two purposes:

1. Word boundary locking
2. Scrambler reset

Idle Packet

The Idle Packet is used to fill the link when there are no requests from the communication channels. It has a variable length limited to a maximum of 64 words. The shortest Idle Packet does not contain any stuffed data and consists only of a two or four symbol header.

Video Packet

Three types of video packets are used for video data transmission: color (Type 1), color with HVD (Type 2), and HVD (Type 3). Packet Types 1 and 2 carry 18 video pixels packed back-to-back with the payload; packet Type 3 carry only 18 pixels of HVD. Packet CRC for all video packet types is disabled by default.

There are three video transmission modes: non-HV-encoding, HV-encoding with all data, and HV-encoding with active data.

In non-HV-encoded mode, the video channel only uses Type 2 data packets.

HV-encoding with all data mode uses both Type 1 and 2 data packets. Type 1 is transmitted when there is no change in HVD information; otherwise, Type 2 is transmitted.

HV-encoding with active data uses all three data packet types. Type 1 packets are transmitted during active data periods (i.e., DE=1) when HVD is not transitioning. Type 2 is transmitted if there is at least one HVD transition in the 18 pixels carried by the packet. Type 3 packets are transmitted during video blanking periods (i.e., DE=0) when there are no HVD changes. Note that the color data during these video blanking periods are not transmitted. Additionally, although the HVD data are carried in these packets, they are not used because the receiver acknowledges there is no change in HVD values when a Type 3 packet is received. Type 3 packets are necessary for video clock regeneration in the receiver to properly track the number of pixel clocks to be generated at the output.

Scrambling

The GMSL2 scrambling randomizes all elements of the data stream excluding special 9b10b symbols. Scrambling is needed for proper clock and data recovery (CDR) by the receiver and EMI reduction. The GMSL2 uses a synchronous scrambling method that involves synchronizing the transmitter and receiver via the sync packet. This approach eliminates the error multiplication of self-synchronizing scramblers. The synchronizer generator polynomial is $58x + 39x + 1$.

PHY Control

The GMSL2 link includes an echo cancellation circuit in both the serializer and deserializer to use the high-speed serial link in full-duplex mode. Integrated PHY controller state machines are activated and perform a calibration sequence after the devices are powered up.

Scheduler/Arbiter

The scheduler will transmit packets with high priority values before lower priority values. Each communication transmit adapter sets a priority for the packet request before transmitting data to the remote side. The priority value is 2 bits, allowing for four different settings: 0=Low, 1=Normal, 2=High, and 3=Urgent. The scheduler, provided sufficient link bandwidth, chooses to transmit the packet with the highest priority among the pending active requests. Priority levels become more important as link bandwidth becomes scarce, and prioritization should be assigned accordingly.

In most cases, each transmitter adapter should use the normal priority setting (priority=1) to allow the scheduler to choose the transmission schedule based on recent bandwidth usage. Packet priority can be increased if packets require low latency or have waited for a length of time. For example, packets with maximum latency requirements can have increased priority if the packet request has not been serviced until half of the maximum latency requirement has elapsed. This can also be used for communications channels with continuous data flow (e.g., Video). Priority can be increased when the transmit adapter data buffer approaches overflow. Conversely, register configuration allow overriding the priority settings of each channel. This option is useful if the host μ C wants to prioritize one channel over the others.

NOTE: Communications channels with very relaxed latency requirements can use the low priority setting (priority=0). These low priority packets will not be serviced by the scheduler if there are any pending requests of a higher priority setting. In this arrangement, link bandwidth assignment can reach the theoretical maximum for Video. Low priority packets can then be transmitted during video horizontal blanking time, during which the Video channel bandwidth usage drops considerably.

Bandwidth Sharing

Link bandwidth is shared flexibly between different communication channels requesting the link for packet transmissions. This flexibility comes from packet-based transmission format and dynamic bandwidth allocation: if a certain channel is not active, it does not consume any link bandwidth, leaving the full link bandwidth available for all active communication channels to share. The packet-based protocol allows fulfilling this sharing requirement. The maximum packet size is limited to 64 words in order to prevent one single channel from monopolizing the link bandwidth and to ensure other channels are served. Other bandwidth sharing mechanisms are discussed below.

A bandwidth sharing scheme is employed in cases of multiple pending requests of the same priority setting to avoid creating buffer overflows. Without bandwidth sharing, a burst data request from a channel could cause buffer overflow in other communications channels on the link. Bandwidth sharing, however, considers predefined bandwidth share ratios and recent bandwidth usage averages of each type of communication channel to avoid buffer overflows and ensure all channels are served.

The arbiter continuously measures the bandwidth usage of each channel and filters the data according to analysis of the moving averages. This data is then compared to assigned bandwidth share ratios. In order to ensure link bandwidth parity, the arbiter takes this recent bandwidth usage information to decide which channel is allocated to use the link for each new packet transmission request. Consider a link allocated

for either Video or SPI communications where Video is assigned 90% link share and SPI is given 5% share. A burst SPI transmission request on this link could result in Video overflow buffering if the SPI bandwidth share were to exceed 10% of the total. In order to avoid Video buffering, SPI is allotted no more than 10% of the link to preserve the remaining 90% for Video communications.

Latency

Latency requirements of each channel are satisfied when link bandwidth is enough to carry packets of all channels. Each communication channel will request to use the link with a certain priority setting based on the latency requirement of that packet. Scheduler will transmit the packets with high priority values before the packets with lower priority values.

CSI-2 Input

The CSI-2 input can be configured for 1, 2, 3 or 4 data lanes.

Deskew of data lanes relative to the clock is enabled for bit rates >1.5Gbps and optional for bit rates ≤1.5Gbps.

To ease PCB layout, swapping of data lanes and the polarity of differential signals within a lane is supported. For example, data lane 3 could be swapped with data lane 2 (lane swap), and the inverting and non-inverting output of data lane 1 could be swapped (polarity swap).

The virtual channel and/or data type received from the CSI-2 source can be passed-through or re-assigned.

For proper start-up at power-up or resuming from sleep or power-down, the clock lane must see the LP-to-HS transition (driven by the source).

CSI-2 Virtual Channel and Data Type Interleaving

Parallel Input

A parallel input with inputs for 14-bit data, HSYNC, VSYNC and pixel clock is available. The parallel input is selected by CFG1 at power-up, or by programming registers. Certain MFP pins and CSI-2 inputs are programmed to function as parallel inputs. The number of data inputs is programmable (i.e., parallel data width can be programmed to 12-bits). The maximum pixel clock frequency is 150MHz. The active (latch) edge of the pixel clock is programmable. The Input Timing Generator can be used to alter timing relationship of the input video sync signals or to derive all sync signals from a single VSYNC input.

CRC Generation and Checking

Every packet (excluding Idle and Acknowledge packets) can be protected by a 16-bit packet Cyclic Redundancy Check (CRC). Each packet type can be individually configured to enable or disable packet CRC. By default, all low-bandwidth channels have packet CRC (Note: Since Acknowledge packets contain no data and the header is repeated, CRC is unnecessary). Although the video channel disables packet CRC in default settings in order to maximize usable bandwidth, Video Line CRC, a 32-bit code at the end of each DE or HS pulse, is enabled by default to provide data protection.

The 16-bit packet CRC generator polynomial is: $x^{16} + x^{15} + x^2 + 1$.

Control Channel Retransmission on Error

Automatic Repeat Request / Automatic Retransmission (ARQ)

Communications channels with control data and low bandwidth require the highest data integrity protection. An automatic packet retransmission method, ARQ (Automatic Repeat Request), is employed here. ARQ works in conjunction with 16-bit packet CRC to detect whether packets are received without error or not.

ARQ transmit and receive blocks are responsible for enabling the bridge between adapters and the scheduler or de-packetizer. Each adapter has a different ARQ path with its own packet length.

Packets from adapters are appended with a 2-bit sequence number at the transmit side; acknowledge is sent from the receiver side upon successful receipt of each data packet. These packets are stored on the transmit side until they are acknowledged. If the acknowledge does not arrive in a predetermined interval or the sequence number of the acknowledge does not match the expected value, the packet waiting at the top of the queue is automatically retransmitted.

The Acknowledge packet uses the same header field as low bandwidth packets, but begins with a different special symbol to distinguish it from regular data packets. This simplified format keeps retransmission exchanges independent from the communication channel. Note that this smaller packet format contains no data, obviating the need for full 16-bit CRC. Instead, the Header symbol is sent twice in the packet and checked against each other to ensure a match. The Acknowledge packets also include a 2-bit sequence number that is the same sequence number of the correctly received data packet. The Data packet transmitter keeps track of which packets have been acknowledged.

Go-Back-N

The ARQ mechanism operates using the 'Go-Back-N' principle, where $N=3$ for a 2-bit sequence number. This three packet sliding window improves bandwidth usage as three packets can be transmitted without receiving an acknowledge.

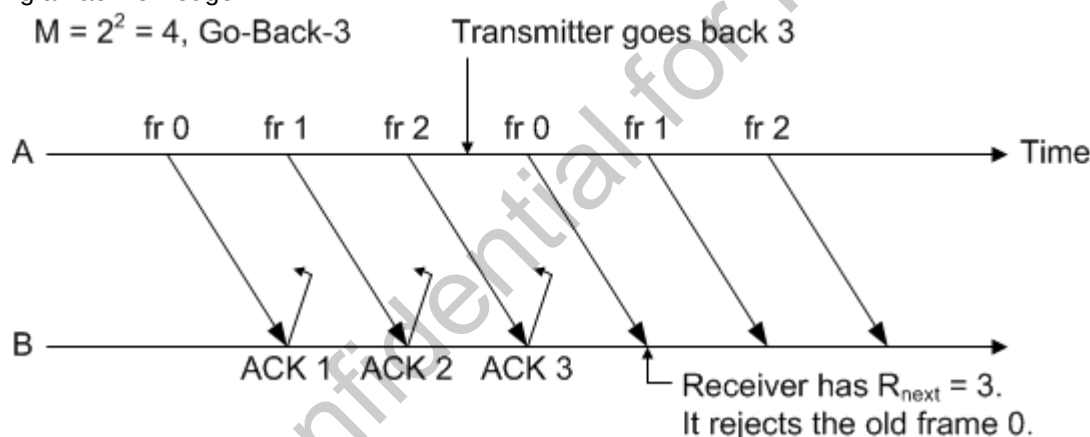


Figure 1 Operation of Go-Back-3 ARQ

First, the transmitter sends up to three packets. After the first packet is sent, the transmitter begins waiting for the acknowledge; if more than one packet is sent, waiting and transmission of the additional packets is handled in parallel. If the expected acknowledge does not arrive, the transmitter goes back three and retransmits the packet. Any retransmission attempt is counted and stored. This retransmission process repeats until the package is either acknowledged or the maximum number of retransmissions is reached. Upon reaching the maximum, the packet is assumed to be acknowledged, it is dropped from the transmission queue, and the send window moves forward to transmit subsequent packets. In cases where the maximum number of retransmissions is met, a register error is flagged and the ERRB pin changes state to signal 'Error Output.'

NOTE: A 3-bit register is allocated to set the maximum number of retransmissions. A packet can be retransmitted up to seven times depending on the register value set in MAX_RT for each communication channel.

The receiver side has a receiver window of size one. Thus, the receiver only waits for one packet with a matching expected sequence number. The receiver passes a packet to the corresponding adapter if the packet is received completely without error and the sequence number matches the expected value. Here,

the receiver also sends an acknowledge packet back to the transmitter (regardless of the received sequence number). In the case of an error or unmatched sequence number, the receiver will drop the received packet.

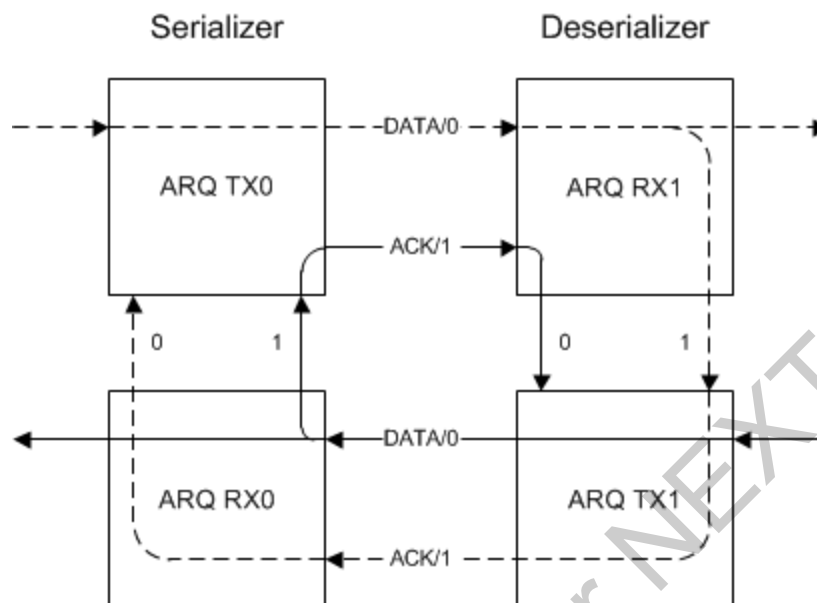


Figure 2 ARQ Path

The dashed line in the ARQ Path figure shows the data flow from the serializer to the deserializer and the subsequent acknowledge from the deserializer to the serializer. The solid line depicts the data flow from the deserializer to the serializer and the subsequent acknowledge from the serializer to the deserializer. The ARQ blocks of both the serializer and the deserializer can operate simultaneously, supporting bidirectional data transfer(s). For example, ARQ TX0 (in the serializer) can transmit either data or an acknowledge of data received from the deserializer. The same function is performed by ARQ TX1 in the deserializer. Here, ARQ TX1 can transmit either data or an acknowledge of data received from the serializer. The ARQ RX blocks operate in a similar manner with the corresponding ARQ TX blocks. This coordination facilitates bidirectional operation.

For details, see *Specifications for Camera Serial Interface 2 (CSI-2) V1.3*.

SPI

The 4-wire SPI blocks of the serializer and deserializer can operate as either an SPI master or slave, enabling communication between a variety of SPI devices. When the serializer and deserializer are connected via the GMSL2 link, the bridged system effectively contains two independent SPI buses.

NOTE: Only one SPI master and one SPI slave can be enabled on the GMSL2 link at any one time.

To illustrate the SPI bridge, consider a GSML2 serializer/deserializer system which has the serializer SPI block configured as a slave and the deserializer SPI block configured as a master. Throughout this section, 'near-side' SPI bus will refer to an external SPI master connected to the internal SPI slave (of the serializer) and 'far-side' SPI bus will refer to an external SPI slave connected to the internal SPI master (of the deserializer). The internal master is capable of communicating with up to two external SPI slave devices. Note that these SPI block configurations are not dynamic; configuration is determined on the hardware level and registers must be set appropriately. Also, there is no support for sub-byte transfers.

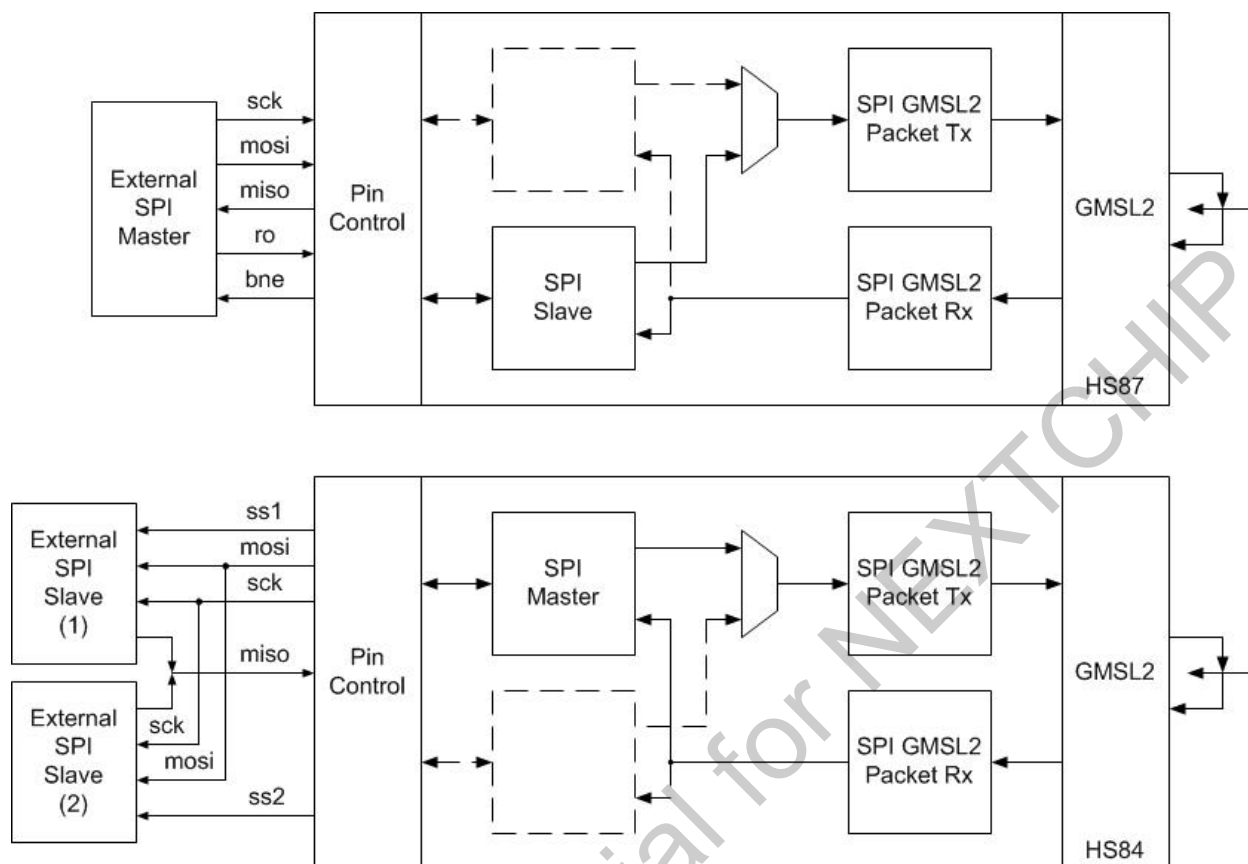


Figure 3 SPI Bridge: SPI Master <-> HS87/HS84 <-> SPI Slave(s)

The near-side SPI bus connects an external master and internal slave in half-duplex mode. This SPI bus supports SPI Mode 0 only. The internal slave uses SCK generated by the external master. Communication on this bus relies on the state of the RO (Read Only) pin to dictate the direction of data movement. The RO signal is an asynchronous input available as a secondary function of a GPIO pin. When RO=0, data is transmitted between the master and slave via MOSI (i.e., writing). When RO=1, data is transmitted between the slave and the master via MISO (i.e., reading). Toggling the RO signal resets the SPI slave logic. In general, the RO signal should be stable (i.e., low or high) for at least one SCK period prior to initiating an SPI transaction (i.e., pulse SCK eight times). After an SPI transaction has concluded, RO should remain stable for at least one SCK period before changing states or toggling. An additional pin, BNE (Buffer Not Empty), is a secondary function of a GPIO pin which is asserted to indicate data is available to be read by the external master. This is an asynchronous output that is not synchronized to the near-side bus.

The far-side SPI bus connects the internal master to external slave(s) and provides full-duplex synchronous serial communication. This SPI bus supports SPI Modes 0 or 3. The internal master uses a divided-down GMSL2 Tx/Rx clock to generate the SCK signal. This bus always operates in full-duplex mode: every byte transmitted to the slave via MOSI results in a byte received by the master via MISO. Two SS (Slave Select) lines with programmable polarity are configurable as secondary functions of GPIO pins. When communicating with SPI slave 1, the SS1 signal is asserted and the SS2 signal is de-asserted; the pattern is reversed to communicate with SPI slave 2. Note that whether transmit data is used by the slave or receive data from the slave is useful is protocol specific.

Initiating SPI Slave Activity Through GMSL2 Bridge

SPI communication is initiated on the near-side SPI bus when the external SPI master sets RO=0 for half-duplex SPI Tx operation. The external SPI master then transmits one or more bytes to the internal SPI slave which are buffered in the packet Tx block. A forward GMSL2 request is then issued. When the

forward GMSL2 request is granted, all available bytes from the near-side packet Tx are sent to the far-side packet Rx block. Here, for each byte in the packet Rx buffer, the SPI internal master performs a full-duplex byte transfer with the external slave on the far-side. A reverse GMSL2 request is then issued. When the reverse GMSL2 request is granted, all available bytes from the far-side packet Tx are sent to the near-side packet Rx block. On the near-side, the BNE pin is de-asserted during read and re-asserted if one or more bytes of data remain and are available for reading by the external SPI master. Note that the host will wait one SCK period after a byte transfer before testing the BNE signal.

Results from SPI Slave Activity

BNE de-assertion or re-assertion creates an edge for the host to use as a rising-edge interrupt detector. Upon edge detection, the external SPI master sets RO=1 for half-duplex SPI Rx operation. With the direction of communication established, the external SPI master then receives one byte from the slave pulled from the buffer in the packet Rx block. The BNE pin is de-asserted during the read operation and subsequently re-asserted if one or more bytes of data remain and are available for reading.

Avoiding Buffer Overflow

All SPI GMSL2 packet Tx/Rx blocks contain a 16B buffer configured as a FIFO. The amount of bytes-in-transit (BIT) should be limited to one buffer's worth (i.e., 16B) of data due to quantization of data throughout the GMSL2 system. The near-side master computes BIT by subtracting the amount of valid bytes read from the amount of valid bytes sent. Note that valid bytes are data ultimately sent or received from the far-side slave.

In-Band Control Mechanism

When RO=1, the near-side slave monitors MOSI for valid control values. The following values are supported:

- 0xA0–0xA3: Used to control value in target field of GMSL2 header
 - This is used in systems with multiple deserializers on the far-side. Use only when BIT=0.
- 0xA4: Assert SS0 of currently targeted far-side device
- 0xA5: Assert SS1 of currently targeted far-side device
- 0xA6: De-assert all SS's of currently targeted far-side device
- 0xA7: Send a data byte (all 1's) to far-side
 - This is used to initiate another far-side slave read while reading a valid byte from the near-side FIFO. Note: This is a valid data byte sent; if used in conjunction while reading a valid byte, the amount of data in transit remains unchanged.
- All other values are silently ignored

NOTE: The SS Control values (i.e., 0xA4-0xA6) result in a byte in the near-side Tx buffer and far-side Rx buffer, but is stripped by the far-side master. Typically, no more than two SS control bytes will be in transit simultaneously (e.g., SS de-assert and SSx assert); this can be accounted for by limiting data in transit to 14B.

Reads from the near-side slave should be avoided when BNE=0 unless the BIT is zero (i.e., no valid data is expected) to avoid discarding a valid data byte potentially arriving during the read.

- 0xA0-0xA3 must be done with BIT=0
- 0xA4-0xA6 will typically be done with BIT=0 or BNE=1
- 0xA7 will typically be done with BNE=1

Multiple External SPI Slaves and the GMSL2 Bridge

The following describes a scenario with two external SPI slave devices on the far-side bus of a serializer/deserializer link.

The external SPI master sets RO=1 for half-duplex SPI Rx operation on the near-side bus. The external master sends a valid control byte to the internal SPI slave on the MOSI line. If BNE is true, the master

must process received MISO data as valid; if BNE is false, the master will discard MISO data. Target control bytes are intercepted by the packet Tx block and append the value in the target field of the GMSL2 header. Slave select control bytes are intercepted by the far-side SPI master on the far-side bus and control state of SS outputs.

SPI Bridge Control and Status Registers

Topology

- Far-side/Near-side: Used to select faster or Tx/Rx GMSL2 clocks for SPI logic. Note, this is the video FS/NS, which is independent of the SPI bridge FS/NS.
- Ignore ID: Instruct packet Rx block to ignore header ID and accept all SPI bridge packets.
- Local ID: ID of local target, used when Ignore ID=0 to filter SPI bridge packets.

GMSL2

- Global N: Maximum packet size for Tx Packets
- Local N: Maximum packet size for SPI Bridge Packets. Superseded by Global N if smaller.
- Base Priority: Starting Request Priority for SPI Bridge Packets. Incremented by 1 if Tx FIFO is greater than half full.
- Tx Overflow: Flag indicating Tx Packet FIFO overflow.
- Rx Overflow: Flag indicating Rx Packet FIFO overflow.

SPI Master

- Enable: Enable SPI Master function and mux common resources to SPI Master block.
- SS I/O Enable: Individual I/O enables for SS0 and SS1.
- SS Active High: Individual polarity control for SS0 and SS1.
- SPI Mode 3: Selects between Mode 0 and Mode 3 for far-side SPI bus.

SPI Slave

- Enable: Enable SPI Slave function and mux common resources to SPI Slave block
- RO I/O Enable
- RO: Control for RO function if I/O is not enabled
- BNE I/O Enable
- BNE: Status for BNE state if I/O is not enabled
- Byte Count: Number of data bytes available for reading from Rx Packet FIFO

\bit equivalencies

- RO: SPI_6 bit0
- BNE: SPI_6 bit1
- SS1: SPI_6 bit2
- SS2: SPI_6 bit3

Control Channel Interfaces

The control channel is available for the μ C to send and receive control data over the GMSL2 serial link simultaneously with the high-speed data. The μ C controls the link from either the serializer or the deserializer side to support video-display or image-sensing applications.

UART

UART can operate in two different modes in GMSL2: base mode and bypass mode.

UART Base Mode

Base mode is enabled by default at power up. In base mode, the μ C is the host and can access the registers of both the serializer and deserializer from either side of the link using the GMSL2 UART packet

protocol. The μ C can also program the peripherals on the remote side by sending the UART packets to the serializer or deserializer, with the UART packets converted to I²C by the device on the remote side of the link. The μ C communicates with a UART peripheral in base mode (through INTTYPE register settings), using the half-duplex default GMSL2 UART protocol of the serializer/deserializer. The device addresses of the serializer and deserializer in this mode are programmable.

When the peripheral interface is I²C, the serializer/ deserializer converts UART packets to I²C that have device addresses different from those of the serializer or deserializer. The converted I²C bit rate is the same as the original UART bit rate.

[GMSL2 UART Protocol for Base Mode Figure] shows the UART protocol for writing and reading in base mode between the μ C and the serializer/ deserializer.

UART Bypass Mode

In bypass mode, the serializer/deserializer ignore UART commands from the μ C and the μ C communicates with the peripherals directly using its own defined UART protocol. The μ C cannot access the serializer/deserializer's registers in this mode.

UART bypass mode can be enabled by two different methods: with the BYPASS_EN register or the MS pin (shared with GPIO). To control UART bypass mode via the register bit, set the BYPASS_EN register in the remote device first and then in the local device to enable 'soft' bypass mode. Note that with this method, bypass mode remains active only as long as there is UART activity; when there is no UART activity for selected duration (configured by BYPASS_TO register), device exits bypass mode and the bit is automatically cleared. Note, If BYPASS_TO is set to "111", bypass mode is permanent and BYPASS_EN stays high until next power down. To control UART bypass mode with the pin, set MS pin. In order to enable bypass mode with the MS pin:

1. Drive GPIO2 (MS) pin Low.
2. Set GPIO_TX_EN_2 of the local device.
3. Set REM_MS_EN in the remote device.
4. Set LOC_MS_EN in the local devices.
5. Drive GPIO2 (MS) pin High from local side to switch to bypass mode.
6. Perform UART communication with the peripheral UART slave.
7. Drive GPIO2 (MS) pin Low from local side to switch to base mode.
8. Perform UART communication with devices.

NOTE: Steps 4 to 7 can be repeated as many times as required.

When MS is High, UART is in bypass mode and when it is Low, UART is in base mode. See \ref 'Device Addressing' for information regarding control channels and device address.

Bit Rate

The deserializer uses differential line coding to send signals over the reverse channel to the serializer. The bit rate of the control channel is 9.6kbps to 5Mbps in both directions. The serializer and deserializer automatically detect the control-channel bit rate in base mode using the sync frame at the start of each packet. Packet bit-rate changes can be made in steps of up to 3.5 times higher or lower than the previous bit rate. See the \ref 'Changing the Clock Frequency' section for more information. The μ C can begin sending packets with the new bit rate provided that it has waited for 32 bit times (as measured by the slower of the old and new bit rates). In bypass mode, the bit rate cannot be changed. The last bit rate that was used in base mode before entering the bypass mode is used in bypass mode.

As shown in [Format Conversion Between GMSL UART and I2C with Register Address Figure], the remote-side device converts packets going to or coming from the peripherals from UART format to I2C format and vice versa. The remote device removes the byte number count and adds or receives the ACK between the data bytes of I2C. The I2C bit rate is the same as the UART bit rate.

Frame Format

Regular UART frames with an even parity bit are used to carry one byte of data each. A frame consists of a low start bit followed by eight data bits, a parity bit, and is finished with a high stop bit. The parity bit is high if the number of ones in 8 bit data is odd, otherwise it is low. There must be at least one high stop bit. If next frame is in the same packet, there can be at most four high bits from the end of the stop bit to the beginning of the next start bit. Note that in the case of a parity bit error, the packet, starting from the frame with the error, will be discarded.

The start of each frame is always a high-to-low transition (i.e., the stop bit is high and the start bit is low). The phase of the internal UART bit clock is adjusted using the start bit of each frame. The framer calibrates the length of one UART bit in terms of the internal oscillator clock using the synchronization frame (i.e., the first frame of a UART packet transmission).

In bypass mode, the parity bit is enabled by default, but the frames are not checked for parity errors. Either even or odd parity can be used. The parity bit is passed along with UART data transmissions; the receiver performs error checking. The parity bit can optionally be disabled in bypass mode by setting `BYPASS_DIS_PAR` to 1 before entering bypass mode. Note that the bit rate in bypass mode must be the same bit rate last used in base mode.

Synchronization Frame

The serializer/deserializer must calibrate internal bit length counters with the UART bit rate for proper recovery of UART frames. A sync frame (a regular UART frame with the value 0x79) is sent as the first frame of each data packet from the μ C and is used to calibrate the bit length in terms of the device's internal 150MHz clock. Sync frames must be properly detected before the subsequent frames of the packet can be correctly received. When the line stays high for at least 32 bits, the packet boundary is reset and the framer begins waiting for the next sync frame.

Acknowledge Frame

When a packet is successfully received, the addressed device responds with an acknowledge frame to inform the μ C that no errors were detected in the transmitted packet and it was recognized as valid. This is sent after the last bit of a successfully recognized packet has been received. The acknowledge frame is a regular UART frame with the value 0xC3.

Data written to the serializer/deserializer registers do not take effect until after the acknowledge byte is sent. This allows the μ C to verify that write commands are received without error, even if the result of the write command directly affects the serial link. If the GPI or MS inputs of the device toggle while there is control channel communication, or if a line fault occurs, the control channel communication will be corrupted. In the event of a missed or delayed acknowledge (~1ms due to control channel timeout), the μ C should assume there was an error in the packet transmission or response. In base mode, the μ C must keep the UART Tx/Rx lines high no more than 4 bit times between bytes in a packet. Keep the UART Tx/Rx lines high for at least 16 bit times before starting to send a new packet.

Write Packet

Write packets consist of a five byte packet header followed by one or more data bytes. A packet is recognized as a write packet when the LSB of the device address frame is 0. The addressed device responds with an acknowledge frame if no errors were detected while receiving the write packet and write packet is valid. Byte count indicates the number of data bytes to be written (N) and it cannot be 0 [REFERS TO DIGAGRAM].

Read Packet

Read packet consists of five bytes. The LSB of the device address frame is 1 for read packets. If no errors were detected while receiving the read packet and the packet is valid, the addressed device responds with an acknowledge frame followed by one or more data bytes. Byte count indicates the number of data bytes to be read (N) and it cannot be 0 [REFERS TO DIGAGRAM].

I²C

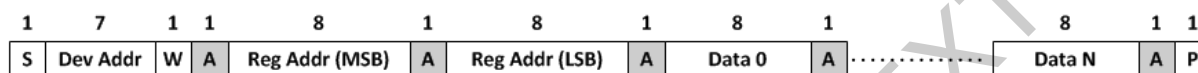
I²C Internal Register Access

Both the serializer and deserializer have an internal I²C slave for internal register access. Internal registers are written and read according to the I²C protocol using packet formats described below. Register addresses are 16-bits wide. Single or multiple data bytes can be written or read (by address auto-increments).

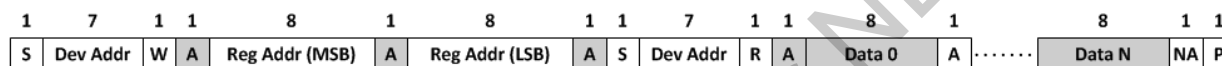
Device Addressing

Each device address is set to one of eight different 7 bit addresses according to the voltage level of ADD2, ADD1, and ADD0 pins at power up. The serializer device address is set to {1000,ADD[2:0]}; the deserializer device address is set to {1001,ADD[2:0]}. Note that device addresses can be changed after power up by writing to the DEV_ADDR register.

Write Packet Format



Read Packet Format



I²C over the GMSL2 Link

The I²C channel connects the serializer and deserializer I²C interfaces together. The interface uses a serial-data line (SDA) and a serial-clock line (SCL) to achieve bidirectional communication between master and slave(s). A μ C master initiates all data transfers to and from the device and generates the SCL clock that synchronizes the data transfer. Functionally, the remote-side internal I²C master mimics the actions of the external I²C master on the local-side; the local-side I²C slave mimics the actions of the I²C slave(s) on the remote-side. In effect, this method logically connects two separate I²C buses—enabling I²C transactions across the link to operate as if performed on the same physical I²C bus.

To illustrate, when an external I²C master performs an I²C transaction on the I²C bus on one side of the link (local-side), events including Start, Stop, Data Bit 0, and Data Bit 1 will then be forwarded to the other side of the link (remote-side) by the internal I²C slave of the local-side chip; the I²C events will subsequently be received by the internal I²C master of the remote-side chip—in turn generating the same I²C events on the remote-side I²C bus. Likewise, the remote-side internal I²C master will send back any I²C events (e.g., Ack bit while writing and Read bits when reading) that are expected to be driven by the I²C slave(s) on the remote-side bus to the local-side internal I²C slave. Note that this arrangement is completely independent of the I²C slave used for internal register access.

The I²C interface uses clock stretching (holding SCL low) to account for timing differences and allow time for data to be forwarded and/or received across the serial link. Note that the external I²C master (and any connected peripherals) must accept clock stretching by the GMSL2 device. The remote side internal I²C master operates according to timing register settings. Configure the I2C_MST_BT setting as close to the bit rate used by the external master as possible.

The SDA and SCL lines operate as both an input and an open-drain output. Pullup resistors are required on SDA and SCL. Each transmission consists of a START condition sent by a master, followed by the device's 7 bit slave address plus a R/W bit, a register address byte, one or more data bytes, and finally a STOP condition.

Applications Configuration

The μ C (external I²C master) can be located on the serializer side (for display applications) or the deserializer side (for camera applications). Dual μ C operations are supported provided that a software arbitration method is used to prevent packet collisions. The serial link assumes that only one μ C is communicating at a time.

Multi-Function Pins (MFP)

Multi-function pins have a default value and are programmable to alternate functions after power-up. See the pin description for default and alternate functions of the MFP pins. Most alternate functions are independently programmable. Some alternate functions are selected as a group (i.e., pass-through I²C channel SDA1/SCL1).

GPIO

The GPIO channel is used to transmit low speed (<100Kbps) signals over the GMSL2 link. MFP pins can be programmed as GPI, GPO (push-pull output) or ODO (open-drain output).

Each GPIO pin can be configured to be an input, output, or input/output by programming the GPIO_TX_EN and GPIO_RX_EN register bits of each GPIO pin. Note that unwanted loop behavior is avoided for pins configured as input/output: when the pin is driven by a transition received from the remote side, the driven GPIO transition is not transmitted back. GPIO pins may alternately be controlled and read solely from registers.

Each GPIO can be programmed for 1M Ω or 25k Ω pull-up or pull-down (or none).

When a GPIO is programmed as GPO, the GPO can be programmed for open-drain or push-pull output.

A GPIO packet has 32 possible GPIO channel IDs. Each GPI is mapped to a channel ID according to the GPIO_TX_ID register. On the receiving end, each GPO outputs the received data with a programmed GPIO channel ID corresponding to the GPIO_RX_ID register for that pin. This provides flexibility in determining which GPIO input drives which GPIO output.

GPIO transmissions are transition based: a GPIO packet is created and transmitted on the GMSL2 link when a rising or falling edge transition is detected at a GPIO pin. Several GPIO transitions at different GPIO pins can be grouped into a single packet. These pin transitions can be transmitted in two different modes: regular and delay-compensated.

The GPIO channel is not bandwidth efficient and should be used for low-speed signals only. Each GPIO transition uses 40-80 bits on the GMSL-2 link for transmit and 40-60 bits on the reverse (due to receive Acknowledge packets). Bandwidth usage values vary based on channel configuration: ARQ enable, CRC enable, and double header enable impact channel bandwidth usage.

Delay in Regular Mode

In regular mode, the value of the transition is transmitted along with the GPIO channel ID. Note that GPIO channel latency is not fixed; the GMSL2 link has variable delay as a result of multiple communication channels sharing the link. A maximum latency limit is established by the GMSL2 Bandwidth Sharing (reference) scheme, however, significant fluctuation remains. Regular GPIO mode should be used with signals invariant to the delay variations (e.g., μ C interrupts).

Delay-compensated Mode

In delay-compensated mode, a time-stamp value is transmitted in addition to the value of the transition and the GPIO channel ID. This time-stamp is a high-resolution value sampled by an internal 600MHz clock that records when the GPIO transition is detected at the input. The remote-side chip uses the time-stamp value to wait and output the GPIO transition after a total fixed delay from the GPIO input transition. This method mitigates possible variable latency issues by making the total GPIO input-to-output delay a

precise, fixed value. Delay-compensated GPIO mode should be used for signals for which the relative timing of rising and falling edges are important (e.g., PWM, camera frame sync, low-speed UART signals).

Frame Sync

In surround-view camera applications a Frame Sync signal is required by the sensor to synchronize the output of a frame with the other cameras in the system. Any GPIO can be configured as a GPO and linked to the Frame Sync signal generated by a GMSL2 deserializer or surround camera ECU.

PRBS

The serializer includes a PRBS generator that works in conjunction with a PRBS checker in the deserializer for bit-error verification of audio and video transmissions.

Video

The video channel of the serializer implements a PRBS pattern generator that operates with bit-error verification in the deserializer to test channel operation. The PRBS generator works with the clock received from the source.

To run the video PRBS test, first enable the PRBS generator on the transmitter side then enable the checker on the receiver side. The PRBS checker will automatically sync itself to the incoming PRBS pattern. If it is unable to sync within a few cycles, it will then assert the PRBS_FAIL register. To stop the PRBS test, first turn the checker off on the receiver side then turn the generator off on the transmitter side.

PRBS errors can be read from VPRBS_ERR register on the deserializer side. Any PRBS error will cause the ERRB pin to go low in the deserializer by default (see VPRBS_ERR_OEN and VPRBS_ERR_FLAG register bits).

NOTE: The video channel shares link bandwidth; it is possible to have a bit error on the link which does not cause a video PRBS error.

Line Fault/Error Indication

Line-Fault Detection

MFP pins can be programmed as line-fault detector inputs (LMN_) and open-drain, multi-purpose line fault/error reporting output (ERRB/LFLT_B). The line-fault detector monitors for serial link line failures, such as short to ground, short to battery, and open link for system fault diagnosis. Figure TBD shows the required external resistor connections. ERB/LFLT_B is low when a line fault is detected and goes high when the line returns to normal. The line-fault type is stored in register 0x0TBD. If serializer and deserializer grounds are different, the link DC voltage during normal operation can vary and cross one of the fault-detection thresholds. Filter ERB/LFLT_B with the μ C to reduce the detector's susceptibility to short-term ground shifts. The fault detector threshold voltages are referenced to MAX9295A ground.

For the fault-detection circuit, select the resistor's power rating to handle a short to the battery. In coax mode, leave the unused line fault inputs unconnected. To detect the short-together case, refer to Application Note 4709: MAX9259 GMSL Line Fault Detection.

Table 19 lists the mapping for line-fault types.

Table 19. Line Fault Mapping

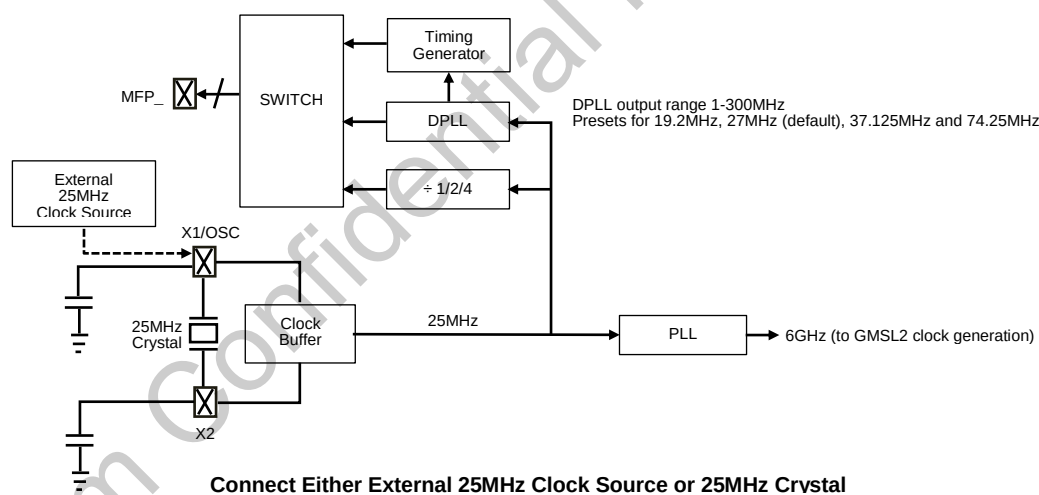
REGISTER ADDRESS	BITS	NAME	VALUE	LINE FAULT TYPE
0xTBD	TBD	LFNEG	00	Negative cable wire shorted to supply voltage
			01	Negative cable wire shorted to ground
			10	Normal operation
			11	Negative cable wire disconnected
	TBD	LFPOS	00	Positive cable wire shorted to supply voltage
			01	Positive cable wire shorted to ground
			10	Normal operation
			11	Positive cable wire disconnected

CSI-2 ECC and Checksum Error Detection

The LFLT/ERRB pin also indicates CSI-2 ECC and Checksum errors and several other configurable D-PHY error indicators. CSI-2 errors generate an interrupt that asserts the ERRB output on the deserializer. Each error that drives LFLT/ERRB has a corresponding status flag in the registers; information regarding LFLT/ERRB assertion is accessible by reading these flags either locally or from the remote (deserializer) side.

Clocking

In GMSL2 mode the 9295A requires an external reference clock source to generate internal device clocks. The external reference can be a 25MHz crystal or oscillator with ± 200 ppm accuracy.

9295A Clock Generation

Connect Either External 25MHz Clock Source or 25MHz Crystal

Crystal Drive Power

100 μ W maximum

Crystal Load Capacitors

TBD

Reference Clock Input Jitter

TBD

Reference Clock Output

25MHz from the XTAL or external reference is available as an undivided or divide by 2 or 4 output by programming the appropriate MFP pin to function as RCLKOUT.

Reference Clock Output Jitter

TBD

Timing Generator (Output)

The video timing generator PLL takes the 25MHz reference as input and outputs a frequency up to 300MHz, and divided values of the reference, on selected MFP pins. These outputs can provide pixel clock and video sync signals to a sensor or ISP.

Timing Generator (Input)

The MAX9295A includes a programmable video timing generator to generate or retime input video sync signals. The timing generator can modify source (i.e., sensor or ISP) input timing, filter out glitches in the sync signals, or to reduce the number of input sync signals. Each sync signal can be individually retimed or left unmodified. Several registers determine the length of the timing parameters (in pixel clock cycles) shown in Figure TBD. Timing parameters include high/low period length, line count, and delay from the input VS signal.

The timing generator uses three different trigger modes: tracking, single trigger, and auto-run.

- 1) Tracking locks once three consecutive identical VSYNC signals are received. The tracker then outputs the same VSYNC signal, filtering any glitches on the input VSYNC. The tracker attempts to relock if three consecutive VSYNC input waveforms do not match the locked signal.
- 2) Single trigger generates one frame for each VSYNC.
- 3) Auto-run generates a new frame at the rate determined by the VSYNC high/low period. If a new VSYNC signal appears before a frame is complete in either single trigger or auto-run modes, a new frame immediately starts, cutting the previous frame short.

Spread-Spectrum Clocking

Spread-spectrum clocking is used to mitigate electromagnetic interference emitted from the devices. Narrow frequency peaks of the clock signal are reduced by spreading the signal over a wider bandwidth. The 25MHz external reference crystal/oscillator is used to generate an internal 6GHz clock. This internal clock is split into parallel signals: one signal is divided by 10, creating a 600MHz clock, and sent to the digital core while a second is divided by 20 to create a 300MHz clock used for spread-spectrum clocking. Digital phase commands are generated on the 300MHz domain to modulate the undivided 6GHz clock used by the Tx/Rx path. This center spreading uses a 25KHz modulation frequency with a sawtooth profile that extends ± 2500 ppm from the center frequency.

Supply sequencing

There is no required sequence to bring up the MAX9295A power supplies. An on-chip power management block manages the power domains during power-up.

However, when a PMIC has the ability to sequence, the following sequence is recommended:

First: VDD18

Second: VDDIO

Third: VDD

Preliminary ramp rate recommendation: $1\mu\text{sec} < \text{ramp rate} < 10 \text{ msec}$

Sleep Mode

To reduce power consumption, the MAX9295A can be put into sleep mode. In sleep mode, all registers keep their programmed values, and all functions in the device are powered down except for the wake-up detectors on the local control interface, and the serial link. Any activity seen by the wake-up detectors temporarily turns on the control-channel interface. During this time, a microcontroller can command the device to exit sleep mode. See the Shutdown/Sleep Modes section.

CFG Latch at Power-up Pins

At power-up or after reset, the voltage at the CFG pins (set by the voltage divider) is sampled. The sampled level is used to set the initial value of certain registers.

Figure TBD. Configuration Pin Circuit

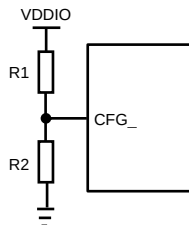


Table 1 - CFG0 Input Map

CFG0 INPUT VOLTAGE (note a, b) (percent of VDDIO)			MAPPED CONFIGURATION (note c)		
MIN	TYP	MAX	I2CSEL	ADDR[2:0]	DEVICE ADDRESS
0.0%	11.02%	13.37% - 17mV	UART (0)	000	0x80
13.37% + 17mV	15.72%	18.21% - 17mV	I2C (1)	000	0x80
18.21% + 17mV	20.70%	23.23% - 17mV	UART (0)	001	0x84
23.23% + 17mV	25.76%	27.97% - 17mV	I2C (1)	001	0x84
27.97% + 17mV	30.17%	32.79% - 17mV	UART (0)	010	0x88
32.79% + 17mV	35.40%	38.00% - 17mV	I2C (1)	010	0x88
38.00% + 17mV	40.60%	42.88% - 17mV	UART (0)	011	0xC0
42.88% + 17mV	45.16%	47.58% - 17mV	I2C (1)	011	0xC0
47.58% + 17mV	50.00%	52.42% - 17mV	UART (0)	100	0xC4
52.42% + 17mV	54.84%	57.12% - 17mV	I2C (1)	100	0xC4
57.12% + 17mV	59.40%	62.00% - 17mV	UART (0)	101	0xC8
62.00% + 17mV	64.60%	67.21% - 17mV	I2C (1)	101	0xC8
67.21% + 17mV	69.83%	72.03% - 17mV	UART (0)	110	0x40
72.03% + 17mV	74.24%	76.77% - 17mV	I2C (1)	110	0x40
76.77% + 17mV	79.30%	81.79% - 17mV	UART (0)	111	0x44
81.79% + 17mV	84.28%	86.63% - 17mV	I2C (1)	111	0x44

Notes:

- Resistor divider tolerance, VDDIO supply ripple and external loading must not cause the CFG0 input voltage to exceed the maximum or minimum limits.
- Other than the CFG0 input resistor divider, any load on CFG0 must be $\geq 25 \times (R1 + R2)$.
- I2CSEL: I2C or UART interface for SDA_RX and SCL_TX
ADDR[2:0]: device address register bits
DEVICE ADDRESS: device address

Table 2 – CFG1 Input Map

CFG1 INPUT VOLTAGE (note a, b) (percent of VDDIO)			MAPPED CONFIGURATION (note c)			
MIN	TYP	MAX	CXTP	HIM/ GMSL2 RATE	PAR/CSI-2	GMSL1/ GMSL2
0.0%	11.02%	13.37% - 17mV	STP (0)	HIM Disabled (0)	CSI-2 (0)	GMSL1 (0)
13.37% + 17mV	15.72%	18.21% - 17mV	STP (0)	6Gbps GMSL2 (0)	CSI-2 (0)	GMSL2 (1)
18.21% + 17mV	20.70%	23.23% - 17mV	STP (0)	HIM Disabled (0)	Parallel (1)	GMSL1 (0)
23.23% + 17mV	25.76%	27.97% - 17mV	STP (0)	6Gbps GMSL2 (0)	Parallel (1)	GMSL2 (1)
27.97% + 17mV	30.17%	32.79% - 17mV	STP (0)	HIM Enabled (1)	CSI-2 (0)	GMSL1 (0)
32.79% + 17mV	35.40%	38.00% - 17mV	STP (0)	3Gbps GMSL2 (1)	CSI-2 (0)	GMSL2 (1)
38.00% + 17mV	40.60%	42.88% - 17mV	STP (0)	HIM Enabled (1)	Parallel (1)	GMSL1 (0)
42.88% + 17mV	45.16%	47.58% - 17mV	STP (0)	3Gbps GMSL2 (1)	Parallel (1)	GMSL2 (1)
47.58% + 17mV	50.00%	52.42% - 17mV	COAX (1)	HIM Disabled (0)	CSI-2 (0)	GMSL1 (0)
52.42% + 17mV	54.84%	57.12% - 17mV	COAX (1)	6Gbps GMSL2 (0)	CSI-2 (0)	GMSL2 (1)
57.12% + 17mV	59.40%	62.00% - 17mV	COAX (1)	HIM Disabled (0)	Parallel (1)	GMSL1 (0)
62.00% + 17mV	64.60%	67.21% - 17mV	COAX (1)	6Gbps GMSL2 (0)	Parallel (1)	GMSL2 (1)
67.21% + 17mV	69.83%	72.03% - 17mV	COAX (1)	HIM Enabled (1)	CSI-2 (0)	GMSL1 (0)
72.03% + 17mV	74.24%	76.77% - 17mV	COAX (1)	3Gbps GMSL2 (1)	CSI-2 (0)	GMSL2 (1)
76.77% + 17mV	79.30%	81.79% - 17mV	COAX (1)	HIM Enabled (1)	Parallel (1)	GMSL1 (0)
81.79% + 17mV	84.28%	86.63% - 17mV	COAX (1)	3Gbps GMSL2 (1)	Parallel (1)	GMSL2 (1)

Notes:

- Resistor divider tolerance, VDDIO supply ripple and external loading must not cause the CFG1 input voltage to exceed the maximum or minimum limit.
- Other than the CFG1 input resistor divider, any load on CFG1 must be $\geq 25 \times (R1 + R2)$.
- CXTP: coax (SIOP) or shielded-twisted-pair (SIOP, SION) serial link.

HIM/GMSL2 RATE:

- HIM applies when GMSL1 operating mode is selected. High Immunity Mode for reverse control channel.
- GMSL2 RATE: applies when GMSL2 operating mode is selected. 3Gbps or 6Gbps GMSL2 serial link bit rate. Control channel rate is 187.5Mbps for both 3Gbps and 6Gbps rate.

PAR/CSI-2: parallel or CSI-2 input.

GMSL1/GMSL2: GMSL1 or GMSL2 operating mode.

Register Table

Revision History

Date	Version	Change
Dec 6, 2016	1.0	None (Initial version)
Dec 19, 2016	1.1	Corrected CFG0 Table
July 21, 2017	1.2	1) Added User-Defined and Generic long packet support. 2) Added watermark insert/detect to block diagram. 3) Increased upper limit ABS MAX a) VDD18 from 1.9V to 2.0V b) VDDIO from 3.6V to 3.9V 4) Increased operating range a) VDD18 from 1.71V/1.89V to 1.7V/1.9V b) VDDIO from 1.71V/1.89V to 1.7V/1.9V 5) Added CNTLx alternate function GPI in GMSL1 mode. 6) Updated AC and DC Electrical Tables